Design and Development of On-Board Diagnostic (OBD) Device for Cars

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Abstract—This paper presents **the design and development of On-Board Diagnostic (OBD) device for cars and its working based on OBD-II standards defined by SAE. This device shows the real-time status of vehicle's systems and sub-systems including engine speed rpm, coolant temperature, pressure etc. and Diagnostic Trouble Codes (DTCs). The software used in this system has been primitively developed which shows the faults in the system. This device helps the user to understand real-time status of vehicle as well as it makes easier to check the malfunctioning in vehicles systems and subsystems by displaying Diagnostic Trouble Codes DTCs). This device doesn't have battery hence it drives power from vehicle battery. The communication between OBD connector and PC is wireless communication.**

Keywords— On-Board Diagnostic (OBD), Diagnostic Trouble Codes (DTCs) , Parameter Id (PID) , Electronic Control Unit (ECU) , Controller Area Network (CAN) , Graphical User Interface (GUI).

I. INTRODUCTION

Now a day, electronic system plays an important role in automotive field. Due to advanced technologies, automotive systems are controlled by electronic system. Hence it is difficult to find out faults occurred in vehicle's systems or subsystems. To know the actual location and the actual fault in systems OBD-II is used. By using OBD-II it is very easy to access this information in a user friendly form to user.

In 1988, the "California Air Resources Board" (CARB) introduced OBD to control or reduce the air pollution caused by traffic. To give information regarding malfunctioning in vehicle's systems and subsystems a Malfunction Indicator Lamp (MIL) is placed at the dashboard of cars.

The first development of On-Board diagnostics specification carried out by Society of Automotive Engineering (SAE) and was named as OBD-I. The purpose of designing OBD was to encourage vehicle manufacturers to design secretion control systems which will be more reliable and eco-friendly. Due to an advancement in technologies, OBD-II was developed which was the improved or advanced version of OBD-I. OBD-II fulfils the drawbacks of OBD-I and also in the span of time ECUs were more capable to provide more diagnostic as well as sensor data to help user to understand the actual position of malfunctioning in vehicles systems. After introducing OBD-II, it was mandatory in all vehicles.

OBD-II is mandatory in all cars which sold after 1996. The Diagnostic Trouble codes given by OBD-II contain different number of characters including alphabets and numbers. These DTCs has 5 characters. Society of Automotive Engineers (SAE) defined some generic DTCs as well as there is a provision to manufactures to make list of their own DTCs and they are known as manufacturer specific DTCs. These codes are very useful to identify and monitor faulty subsystems of Vehicles. DTCs are accessed through OBD kit and displays on primitively developed GUI Display Application. The OBD device is connected to laptop which has primitively developed Graphical User Interface (GUI) which displays the Diagnostic Trouble Codes (DTCs) in user friendly form.

In this system, the interfacing of OBD connector and Laptop is carried out using an automotive grade processor KEAZ128 and HC-05 Bluetooth module. When request of diagnosis is sent to ECU, then it checks the status of vehicle's systems and subsystems if faults or malfunctioning arises then it sends message on CAN bus which includes information of malfunction and its location in the form of DTCs. This information helps the user to resolve fault occurred in vehicle. This device is user-friendly as well as an affordable device and easier to use.

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II. OBD-II

On-Board Diagnostic is one of the computer system equipped in cars which can discover and diagnose problems occurred in vehicle's systems and subsystems with the data reported by the ECUs. OBD-II gives DTCs which have been generated by the computer system while malfunctioning occurred in vehicle as well as gives the real-time status of the systems according to sensors output which are connected to different systems of vehicle. The main advantage of having OBD is that it makes it easier to diagnose faults that occur in vehicle's systems and subsystems. Another main advantage from a sustainability point of view is that vehicle emissions can be reduced by diagnosing and monitoring faults that makes vehicle emission level rise.

OBD-II connector is complaint to J1962 standardized female 16 pin connector. OBD-II port is usually located in the foot well of driver's side of car. Now-a-day it is mandatory in all new cars. While 9 pins of OBD-II are mandated by the standard to use from that 16 pins. SAE J1962 standard defines the pin out of the OBD-II connector as shown in figure 1.

A. Signalling Protocol

The OBD-II interface contains five signaling protocols. But in most of vehicles implement only one signaling protocol from that. Also it is possible to decide which protocol should use depends upon pins available on J1962 (OBD-II) connector. These five signaling protocols are as follows:

1. SAE J1850 PWM (pulse width modulation)
2. SAE J1850 VPW (variable pulse width)
3. ISO 9141-2 (K-LINE or L-LINE)
4. ISO 14320 KWP2000 (Keyword Protocol 2000)
5. ISO 15765 CAN

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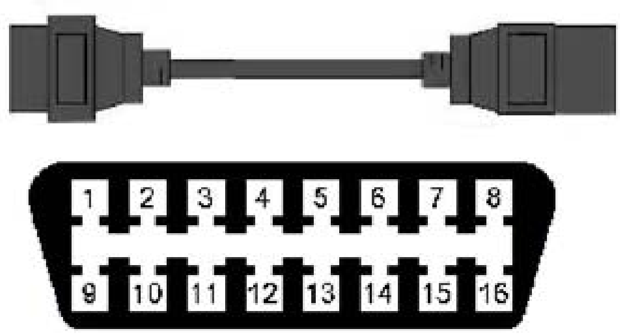

Fig. 1 J1962 (OBD-II) Connector

TABLE I. Pin-out Description

PIN	Description	PIN	Description
1	Manufacturer Specific	9	Manufacturer Specific
2	J1850 Bus+	10	J1850 Bus-
3	Manufacturer Specific	11	Manufacturer Specific
4	Chassis GND	12	Manufacturer Specific
5	Signal GND	13	Manufacturer Specific
6	CAN High	14	CAN Low
7	ISO-9141-2 K-LINE	15	ISO-9141-2 L-LINE
8	Manufacturer Specific	16	+12 V (Battery Power)

B. Diagnostic Trouble Codes (DTCs)

TABLE III. Breakdown Structure of DTCs

Character number	1	2	3	4	5
Description	System	Control	Sub-System	ID	ID
Allowed Values	'P'	'0'	0-F	0-F	0-F
	'B'	'1'			
	'C'	'2'			
	'U'	'3'			

Where,
In systems:
P – Powertrain
C – Chassis
B – Body
U – User Network

In Control:
0 – Generic (ISO specific)
1 – Manufacturer specific
2 and 3 – Reserved

The Diagnostic Trouble codes (DTCs) are used to detect and to monitor malfunction in vehicle's systems and subsystems including their actual location. Diagnostic Trouble Codes (DTCs) has 5 characters including alphabets and numbers and every character has particular meaning. DTCs are defined by SAE are generic DTCs but there are some manufacturer specific DTCs are available i.e., manufacturer can define their own DTC codes. The structure of DTCs defined by SAE has mainly four systems. The Structure of DTCs is as shown in TABLE 2.

C. Vehicle Identification Number (VIN)

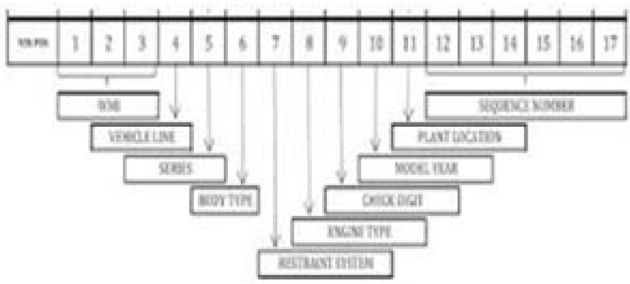

Fig. 2 Breakdown structure of VIN

Vehicle Identification Number (VIN) was originally designed for vehicle identification. It is a unique identification number for vehicle. It is universally accepted 17 digit alphanumeric code (which does not use letters I, O and Q). It is based on the ISO standards 3779 and 3780. VIN is mainly divided in three parts i.e. WMI, VDS, and VIS.

Fig.2 shows the breakdown structure of VIN. Here, 1-3 digits define WMI means World Manufacturer Identifier which describes information of vehicle as manufacturer, country of origin and type of vehicle. WMI is controlled by Society of Automotive Engineers based in U.S. 4-9 digits define VDS means Vehicle Description Section and defines some of the vehicle features such as model style or type of engine. In this, 9^{th} digit is check digit. VDS is managed by manufacturer. 10-17 digits are used for Vehicle Identifier Section which is also managed by manufacturer. 10^{th} digit describes model year defined in ISO 3779. 11^{th} digit describes plant code i.e. country code where actually assembled. And 12^{th}-17^{th}, 5 digits describes the sequential number and these are defined by manufacturer.

D. Controller Area Network (CAN)

Controller Area Network (CAN) is mainly used as OBD-II communication protocol. CAN bus is used as communication bus in vehicles. It is a message based and multi-master protocol. ISO 11898, ISO 15765 and ISO 15031 are standards for diagnostics on CAN. Here in this system I am going to use ISO 15765 standards which used for diagnosis on CAN bus. CAN has been subdivided into different layers of ISO/OSI model and they are the object layer, transfer layer and the physical layer. Basically, CAN is divided into two sub parts such as CAN High (High Speed CAN: speed- 125kbps to 1 Mbps) and CAN Low (Low speed CAN: speed- 5 to 125kbps)

CAN has two types of message identifiers such as 11-bit (Standard) identifier and 29-bit (Extended) identifier. CAN use two logic states such as logic '0' known as 'Dominant' state and logic '1' known as recessive state.

Below Fig. 3 show the MSCAN (Freescale's Scalable Controller Area Network) system which is compatible with Bosch CAN 2.0 protocol. It has error detection and signaling feature.

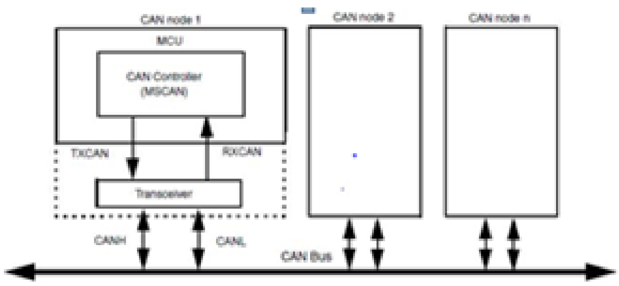

Fig. 3 CAN System

III. PROPOSED METHODOLOGY

This section contains the proposed methodology for designing and development of OBD device for cars. OBD is a device which shows the real time status of vehicle systems as well as malfunctioning occurs in vehicle systems or sub-systems. Hence due to this device fault detecting and monitoring is easy. Here we have designed On-Board Diagnostic device as similar as existing OBD-II device except some extra features. These features are:

1. It has provision that user can add new information which is not available in database.
2. It provides wireless communication between user's laptop or PC and OBD-II connector.
3. It provides GUI along with information stored in database which gives DTCs with its description to user.
4. It is lower costly than existing system hence local user also can be use it.

The main challenging task in this proposed system is to do both Bluetooth and CAN communication simultaneously and synchronously.

A. Proposed System

Proposed system has a main part is of proposed block diagram of On-Board Diagnostic device for cars. In our proposed System we have used ISO-15765 CAN as a signaling protocol of OBD-II. The controller used in this system is of KEAZ128 product family of ARM^R Cortex – M0+ MCUs. HC-05 Bluetooth module is used and it is externally connected to KEAZ128 microcontroller. OBD connector is connected to microcontroller using CAN signaling protocol. A car has OBD-II port to connect OBD-II connector. Proposed block diagram is shown in Fig.3. Block diagram consists of three main blocks such as Laptop or PC, KEAZ128 microcontroller and CAR.

Laptop or PC used in this system has indigenously developed GUI which shows DTCs in a user friendly format which gives information of malfunctioning in vehicle's systems or subsystems.

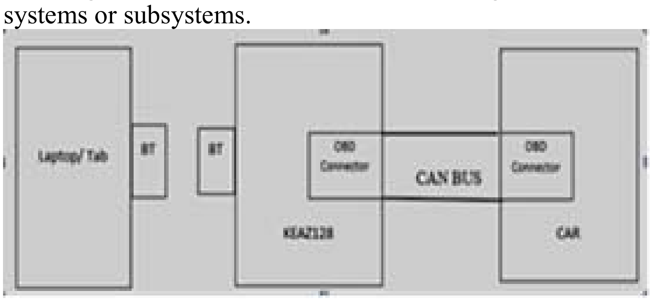

Fig. 4 Block diagram of proposed system (OBD-II)

KEAZ128 microcontroller has inbuilt CAN (MSCAN) protocol. By using this we can connect OBD connector to it.

Vehicle has inbuilt OBD-II port which is 16 pin female connector. Hence by connecting 16 pin male OBD-II connector to it we can establish communication between KEAZ128 microcontroller and ECU of vehicle through CAN bus.

This is the overall methodology to design proposed system.

B. Softwware Algorith for Communication

To establish the communication in this proposed system the following algorithm is used:

1. To check the faults in vehicle send query message from laptop or PC through Bluetooth.
2. Bluetooth module connected to microcontroller will receive query message and decode received message.
3. Microcontroller send decoded message to ECU of vehicle through OBD connector by using CAN protocol.
4. ECU takes action according to received message by CAN and send acknowledgement in the form of DTCs to microcontroller.
5. Microcontroller encode the DTCs and send back to the laptop or PC through Bluetooth.
6. Laptop or PC which has indigenously developed GUI displays received DTCs in user friendly form.

IV. HARDWARE DESIGN

A KEAZ128 microcontroller has been used for interpreting OBD-II protocol. This microcontroller is act as a bridge between Laptop or PC and OBD port in vehicle. KEAZ128 microcontroller is a full featured microcontroller which handles all signalling protocols of OBD-II. It is an automotive grade controller which is specifically used for automotive applications.

The major components of hardware are KEAZ128 microcontroller with inbuilt MSCAN transceiver, OBD-II connector, HC-05 Bluetooth module and Laptop or PC. This is designed as battery free and drives power from vehicle battery. This design is easy to use and cost effective.

A. KEAZ128 microcontroller

KEAZ128 microcontroller is of 32-Bit controller and product family of ARM^R Cortex –M0+ MCUs. It supports 2.7 to 5V power supply and has 48 MHz CPU frequency. It has some in-built systems such as System integration module (SIM), Power management and mode controllers (PMC),

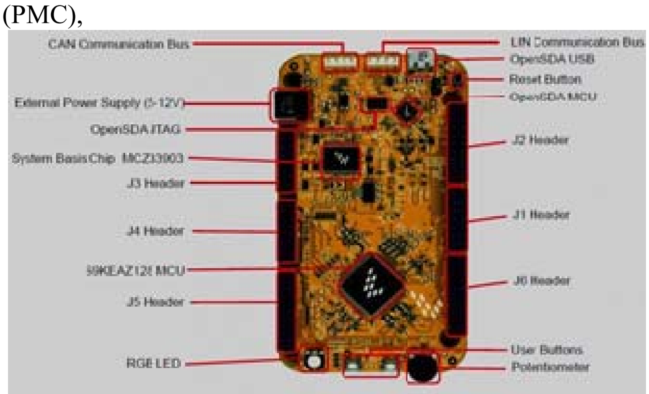

Fig. 5 KEAZ128 microcontroller

Miscellaneous control module (MCM), Bit manipulation engine (BME), Peripheral Bridge (AIPS), Watchdog (WDOG). It has inbuilt timers such as One 6-channel Flex Timer (FTM) with full function, Two 2-channel FTMs with basic TPM function, 2-channel periodic interrupt timer (PIT), Real time clock (RTC), One pulse width timer (PWT), System tick timer (SysTick) and communication modules such as Two 8-bit serial peripheral interfaces (SPI), Two inter-integrated circuit (I2C) modules, Three universal asynchronous receiver/transmitter (UART) modules, One Freescale's scalable controller area network (MSCAN). The hardware kit of KEAZ128 is as shown in Fig. 5.

B. Structure of MSCAN in KEAZ128 microcontroller

The MSCAN is Freescale's Scalable Controller Area Network and which compliant to Bosch CAN 2.0. It has two clocks such as Oscillator clock and Bus clock. It has two external signals, TXCAN and RXCAN. It has low pass filter used for reconstruction of signal. A block diagram of MSCAN is shown in Fig. 6. It has five receive buffers with FIFO scheme.

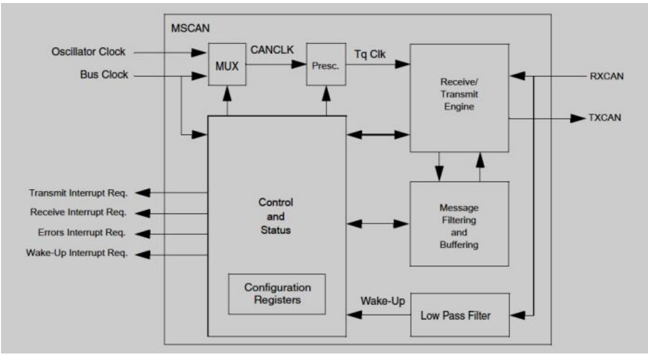

Fig. 6 MSCAN Block Diagram

V. SOFTWARE DESIGN

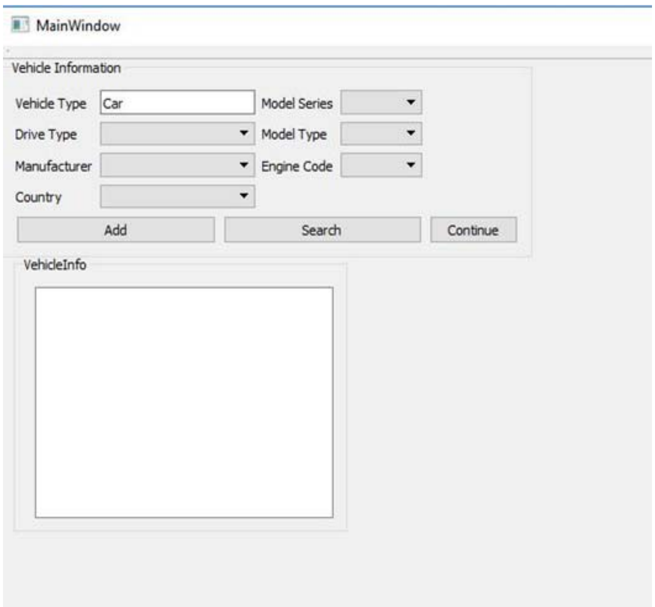

Fig. 7 Main Application Window

A Graphical User Interface (GUI) has been developed to make this system and software user friendly in QT Creator using C language. Using this software we can access different PIDs, real-time status of vehicle and DTCs. It shows DTCs with its specific description which is stored in database. Database stores DTCs with their detailed description as well as information of vehicle. Database has been designed and maintained in SQLite Studio and it is linked with GUI. The main Application window of software is shown in Fig. 7.

VI. CONCLUSION

This system is used to detect and to monitor malfunctioning of vehicle systems and subsystems. This device can work on all models of cars based on OBD-II standards. It shows DTCs with its detailed information in a user friendly form. This is battery less system. It is reliable and cost effective as well as easy to use. It has also given provision to user that user can add new data into database which is not available in database. Hence basically it is user friendly system.

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